Rahul Rajeev, Ph.D.

MATERIALS SCIENTIST & ENGINEER

Materials Processing & Scale-Up | Characterization & Reliability | Electrochemistry
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PROFESSIONAL SUMMARY

Ph.D. materials scientist and engineer with 5+ years of hands-on R&D in materials synthesis, ceramic processing, process scale-up, advanced characterization, and electrochemical testing. Developed a 15-minute powder synthesis route, scaled it from 5 g to 1 kg per batch (~200x), and delivered 25 additively manufactured ceramic tubes to a DOE national laboratory with no observed cracking or leakage. Designed and built custom high-temperature test systems, investigated material degradation, and managed a laboratory of 8–12 researchers, including about $50K of equipment procurement. Four peer-reviewed publications (two first-author) and one invention disclosure.

TECHNICAL SKILLS

Materials Synthesis & Processing: Solution combustion, solid-state and sol-gel synthesis; ceramic powder processing and milling; slurry formulation; direct ink writing (robocasting); slip casting, tape casting, die pressing; reactive sintering; high-temperature, vacuum, and inert-atmosphere furnaces; process optimization and scale-up

Characterization & Testing: XRD/Rietveld refinement (Rigaku, Bruker; GSAS-II); SEM/EDS (Hitachi), FIB, TEM, XPS; optical microscopy; BET and laser-diffraction particle sizing; ICP-OES; TGA/DSC; dilatometry (CTE); FTIR, Raman; Instron flexural, compression, and tensile testing; cross-sectioning and polishing; ImageJ analysis of grain size, porosity, and phase fraction

Electrochemistry & Energy Materials: Electrochemical impedance spectroscopy (Gamry, Solartron, BioLogic; ZView equivalent-circuit fitting); cyclic voltammetry; DC polarization; critical current density; galvanostatic cycling; Li-metal handling and symmetric-cell assembly in an argon glovebox; solid-state electrolytes; interfacial degradation

Process Development & Lab Operations: Planned parameter-matrix experiments; error analysis; equipment installation, calibration, and troubleshooting; thermal cycling and helium leak testing; SOP development; laboratory safety; vendor quoting and procurement; technical reporting

Software & Data Analysis: Python, MATLAB, OriginPro, Microsoft Excel, FactSage, Autodesk Inventor (CAD), GSAS-II, ZView, instrument-control software (Gamry, XRD)

PROFESSIONAL EXPERIENCE

Graduate Research Assistant, Materials R&D | Brinkman Research Group, Clemson University **Jan 2021 – Aug 2026**

Researcher on the DOE ARPA-E-funded GAMOW program in collaboration with Savannah River National Laboratory (SRNL).

- **Synthesis and scale-up:** Developed a solvent-free combustion route that forms cubic Ga-doped LLZO electrolyte powder in 15 minutes at ~90% yield; scaled batch size from 5 g to 1 kg (~200x) and optimized sintering to reach 94.2% relative density and 5.8×10^{-4} S/cm ionic conductivity at 30 °C
- **Ceramic additive manufacturing:** Formulated a 50 wt.% solids ceramic slurry, optimized print geometry and debinding and reactive-sintering schedules (up to 1150 °C), and fabricated closed-end tubes at ~92% relative density; delivered 25 tubes to SRNL with no observed cracking or leakage in gas flow-meter and gas chromatography leak checks
- **Custom test hardware:** Designed and built a multi-sample impedance system (room temperature to 500 °C) and a sealed inert-atmosphere cell for molten Li–Pb exposures (350–450 °C, up to 100 h); tested Li|LLZO|Li symmetric cells by EIS, cyclic voltammetry, critical current density, and 100 h Li stripping/plating
- **Degradation and failure analysis:** Combined XRD/Rietveld, SEM/EDS, XPS, and TEM to compare Ga- and Ta-doped LLZO after molten-metal exposure; identified phase transformation and Pb grain-boundary penetration in Ga-LLZO versus retained cubic structure and a ~2 µm reaction layer in Ta-LLZO
- **Glass seals and reliability:** Screened three SOFC sealants for 350–650 °C service using thermal cycling (~20–30 tests, 300–800 °C), a helium leak-test rig, DSC, CTE mismatch, and adhesion tests; Schott GM31107 was selected
- **Characterization and communication:** Ran XRD on 800+ samples (Rigaku, Bruker), operated Hitachi SEMs independently, presented quarterly updates to DOE ARPA-E for about two years, and mentored undergraduate researchers

Laboratory Manager (concurrent appointment) | Brinkman Research Group, Clemson University **May 2022 – Mar 2024**

- Managed daily operations of an 8–12 person materials research laboratory: equipment scheduling, safety procedures, SOPs, researcher training, and batch records
- Coordinated about $50K of procurement; collected multi-vendor quotes and purchased a Gamry potentiostat, 2–3 box furnaces, furnace tubes and spare parts, and all laboratory supplies

- Installed, calibrated, and troubleshot furnaces and electrochemical instruments; trained 30+ users on XRD and new researchers on furnaces and electrochemical testing

Teaching Assistant | Materials Science and Engineering, Clemson University **Fall 2023, Spring 2024, Fall 2025**

- Taught undergraduate synthesis, fabrication, and characterization labs, including Instron flexural and compression testing, viscometry, and optical microscopy

Undergraduate Research Intern | Brinkman Group, Clemson University **May – Jul 2019**

- Studied CO2 laser sintering of LLZTO thin films, varying laser power, scan speed, and atmosphere; characterized phase and microstructure by XRD and SEM

EDUCATION

Ph.D., Materials Science and Engineering | Clemson University, Clemson, SC **Aug 2026**

B.Tech. (Honours), Materials Science and Engineering | Indian Institute of Technology Gandhinagar, India **Aug 2020**

SELECTED PUBLICATIONS & INTELLECTUAL PROPERTY

- R. Rajeev et al., "Comparative stability of Ga and Ta-doped LLZO in molten Li-Pb: phase evolution and degradation behavior," Journal of Alloys and Compounds 1082 (2026) 191481. doi:10.1016/j.jallcom.2026.191481
- R. Rajeev, A. K. Mishra, K. Brinkman, "Rapid solvent-free synthesis of Ga-doped LLZO," Chemical Communications 61 (2025) 18436–18439. doi:10.1039/D5CC04773K
- Co-author, Materials Letters 375 (2024) 137240, and Journal of Materials Science 58 (2023) 6174–6184 (3D-printed and combustion-synthesized Ga-LLZO)
- Invention disclosure filed with the Clemson University Research Foundation: scalable 3D-printed LLZO ceramic fabrication
- Presentations: ECS Meeting, Seattle (2026, oral); MS&T (2023, poster)

AWARDS & HONORS

MRS/Optica Graduate Poster Competition, 1st Prize (2024, 2026) | Clemson Doctoral Dissertation Completion Grant, $10,000 (2026) | David Jennings Memorial, Kentwool Educational, and Gilbert Robinson Memorial Fellowships